

Aarav J. Samal

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EDUCATION

Purdue University

West Lafayette, IN

B.S. in Computer & Information Technology; John Martinson Honors College

Expected May 2027

- **Relevant Coursework:** Enterprise Data Management, IT Architectures, Cybersecurity Fundamentals

EXPERIENCE

AI Engineer Intern

June 2026 – Aug. 2026

Bristol Myers Squibb

- Embedded directly with a **cross-functional team** of drug-discovery scientists, product, and engineering to scope and ship **30+ features** on a full-stack **Claude Agent SDK** platform (**React, TypeScript, Node.js, Express**) – owning each from ambiguous problem framing through production deployment.
- Designed and deployed a **Claude-driven self-service support system** (custom SSE endpoint, intent detection, autofix pipeline) that let non-technical scientists resolve bugs/feature requests without engineering involvement, cutting manual triage load and turnaround time from pain point to fix.
- Built a **self-learning observability layer** and startup preflight checks that surfaced extraction health across fast/slow pipeline cycles, catching silent runtime failures before data loss and preventing downstream disruption to scientists' workflows.

Mobile & Web App Developer Intern

Aug. 2025 – Oct. 2025

Niramai Thermalytix

- Built a cross-platform mobile app enabling **offline image quality verification**, reducing screening delays for end users operating in low-connectivity clinical environments.
- Integrated a thermal breast cancer screening **AI model** into the web application, developed supporting **Node.js** backend services, and deployed components on **AWS and Azure** for scalable, secure screening workflows.

Co-President

May 2024 – July 2026

PrepTech Foundation

- Co-led a student-run education nonprofit, managing hiring and operations for **30+ student employees** across instruction, outreach, and curriculum delivery.
- Scaled the organization's first session to **100+ students across 10+ states** through targeted outreach and strategic partnerships.

PROJECTS

Britney.ai | *Python, FastAPI, Postgres, Flutter, OpenAI API*

- Built and deployed an autonomous trading agent (**Python, FastAPI, Postgres**) running unattended 24/7 on a **DigitalOcean** droplet, executing risk-tiered portfolio rebalancing across 3 strategies with live discovery, classification, and emergency stop-loss systems – outperforming the market by an average of **7%** in its first month live.
- Integrated **OpenAI GPT-4o-mini** into a tool-calling chat assistant grounded in real trade/cash-ledger data, and shipped a cross-platform **Flutter** frontend backed by a **183-test pytest suite** across 17 files.
- Independently scoped system architecture with no fixed spec – deciding build order, risk tiers, and stop-loss logic from first principles – then validated each design choice against live production trades before scaling further.

Find My College | *Flutter, Dart, Python, Jupyter, MySQL*

- Designed and shipped a mobile application using a predictive **machine learning model** to estimate college acceptance probabilities with **94% accuracy**, giving prospective applicants a data-driven view of their odds before applying.
- Built an interactive **Flutter/Dart** UI and integrated model outputs end-to-end with a MySQL-backed application architecture, from data pipeline to user-facing predictions.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Dart, SQL, Java, C++, C#

Frameworks & Libraries: React, Node.js, Express, Flutter, Vite, Tailwind CSS, Vitest, FastAPI, SQLAlchemy

AI & Agents: Claude Agent SDK, Model Context Protocol (MCP), OpenAI API (GPT-4o-mini), LLM tool use, prompt engineering, semantic similarity search

Cloud & Infrastructure: AWS, Azure, GCP, Docker, GitHub Actions, Git, DigitalOcean, systemd, Linux (Ubuntu)

Tools & APIs: Jira, MySQL, Postgres, SQLite, REST APIs, WebSockets, Server-Sent Events, Yahoo Finance API, Finnhub API

Testing: pytest, Hypothesis (property-based testing), Playwright (E2E)